MTH 101

Calculus & Analytical Geometry

Complete Typed Notes

(Properly Formatted from Handouts)

All 23 Lessons

Includes:

Number Systems • Sets • Inequalities • Absolute Value

Coordinate Plane • Slope • Distance • Circles

Functions • Limits • Continuity • Derivatives

Differentiation Techniques • Applications of Derivatives

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1 Lesson 01: Coordinates, Graphs and Lines

What is Calculus?

Calculus is the study of continuous rates of change of quantities. It is the study of how various quantities change with respect to (w.r.t) other quantities.

Examples:

- How distance changes w.r.t time
- How time changes w.r.t speed
- How water flow changes w.r.t time

1.1 Real Numbers

Type	Definition / Examples
Natural Numbers	$1, 2, 3, 4, 5, \ldots$
Whole Numbers	$0, 1, 2, 3, 4, \ldots$
Integers	$\ldots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots$
Rational Numbers	Ratios of integers p/q (division by zero ruled out)
Irrational Numbers	Cannot be expressed as ratio of integers

Why not divide by 0?

If $x \neq 0$, the equation $x/0 = y \Rightarrow x = 0 \cdot y \Rightarrow x = 0$ is contradictory.
If $x = 0$, the equation is satisfied by any number y, so the ratio does not have a unique value.
For these reasons such symbols are **undefined**.

Irrational Numbers — Historical Note

Pythagoras concluded that the size of a physical quantity must consist of a whole number of units plus some fraction m/n. This idea was shattered in the fifth century B.C. by **Hippasus of Metapontum** who demonstrated the existence of irrational numbers. He proved that $\sqrt{2}$ is irrational.

Examples of irrational numbers: π, $\cos 19°$, $1 + \sqrt{2}$

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Real Numbers

↓

Rational	**Irrational**
(Can be written as p/q)	(Cannot be written as p/q)
↓	(non-terminating,
Integers	
↓	
Whole	non-repeating)
↓	
Natural	
Non-terminating	
repeating decimals	

1.2 Set Theory

Set

The collection of **well-defined objects** is called a set.

Notation:

- $a \in A$: a is an element of set A
- $a \notin A$: a is NOT an element of set A
- $\varnothing$ or $\{\}$: Empty set
- $A \cup B$: Union — all elements of A and B together
- $A \cap B$: Intersection — elements in both A AND B
- $A = B$: A is exactly the same set as B
- $A \subseteq B$: A is a subset of B

Set-Builder Notation:

$$\{x : x \text{ is a real number and } 2 < x < 3\}$$

1.3 Coordinate Line

In the 1600's, analytic geometry was developed by the French mathematician **Descartes**. The key step is to establish a correspondence between real numbers and points on a line.

- The **positive direction** is usually marked with an arrowhead (right side)
- The **negative direction** is the left side
- The **origin** is marked with the number 0
- A **unit of measurement** is chosen

One-to-One Correspondence

Each real number corresponds to a single point and each point corresponds to a single real number. We say that the real numbers and the points on a coordinate line are in **one-to-one correspondence**.

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1.4 Order Properties

For any two real numbers a and b, if $b - a$ is positive, then we say that $b > a$ or that $a < b$.

Property	Rule
(a)	If $a < b$ and $b < c$, then $a < c$
(b)	If $a < b$ and $a + c < b + c$, then $a - c < b - c$
(c)	If $a < b$ and $ac < bc$ when c is positive; $ac > bc$ when c is negative
(d)	If $a < b$ and $c < d$, then $a + c < b + d$
(e)	If a and b are both positive or both negative and $a < b$, then $1/a > 1/b$

1.5 Intervals

Notation	Definition	Type
$[a, b]$	$\{x : a \leq x \leq b\}$	Closed interval
(a, b)	$\{x : a < x < b\}$	Open interval
$[a, b)$	$\{x : a \leq x < b\}$	Half-open
$(a, b]$	$\{x : a < x \leq b\}$	Half-open
$[a, +\infty)$	$\{x : x \geq a\}$	Infinite (closed)
$(a, +\infty)$	$\{x : x > a\}$	Infinite (open)
$(-\infty, b]$	$\{x : x \leq b\}$	Infinite (closed)
$(-\infty, b)$	$\{x : x < b\}$	Infinite (open)
$(-\infty, +\infty)$	$\mathbb{R}$	Both open and closed

Important Note

Infinite intervals of the form $[a, +\infty)$ and $(-\infty, b]$ are considered **closed** because they contain their endpoint. The interval $(-\infty, +\infty)$ has no endpoints; it is regarded to be **both open and closed**.

1.6 Solving Inequalities

Example 4: Solve 3 + 7x less than or equal to 2x minus 9

$$7x \leq 2x - 12 \quad \text{(Subtracting 3)}$$
$$5x \leq -12 \quad \text{(Subtracting } 2x\text{)}$$
$$x \leq -\frac{12}{5} \quad \text{(Dividing by 5)}$$

Solution set: $\left(-\infty, -\frac{12}{5}\right]$

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Example 5: Solve 7 less than or equal to 2 minus 5x less than 9

$$\begin{aligned} 5 \leq -5x < 7 \quad &\text{(Subtracting 2)} \\ -1 \geq x > -\frac{7}{5} \quad &\text{(Dividing by } -5\text{; inequality reversed)} \\ -\frac{7}{5} < x \leq -1 \quad &\text{(Rewriting)} \end{aligned}$$

Solution set: $\left(-\frac{7}{5}, -1\right]$

2 Lesson 02: Absolute Value

Absolute Value

The absolute value or magnitude of a real number a is denoted by $|a|$ and is defined by:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \text{ (non-negative)} \\ -a & \text{if } a < 0 \text{ (negative)} \end{cases}$$

Examples:

$$|5| = 5, \quad \left|-\frac{4}{7}\right| = \frac{4}{7}, \quad |0| = 0$$

Caution

Symbols such as $+a$ and $-a$ are deceptive. If a itself is negative, then $-a$ is positive and $+a$ is negative.

2.1 Solving Absolute Value Equations

Example 1: Solve absolute value of x minus 3 equals 4

$$\begin{aligned} x - 3 = 4 \quad &\text{or} \quad x - 3 = -4 \\ x = 7 \quad &\text{or} \quad x = -1 \end{aligned}$$

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Example 2: Solve absolute value of 3x minus 2 equals absolute value of 5x plus 4

Case 1:

$$\begin{aligned} 3x - 2 &= 5x + 4 \\ -2x &= 6 \\ x &= -3 \end{aligned}$$

Case 2:

$$\begin{aligned} 3x - 2 &= -(5x + 4) \\ 3x - 2 &= -5x - 4 \\ 8x &= -2 \\ x &= -\frac{1}{4} \end{aligned}$$

2.2 Relationship Between Square Roots and Absolute Values

Theorem

For any real number a:

$$\sqrt{a^2} = |a|$$

Proof: Since $a^2 = (+a)^2 = (-a)^2$, the numbers $+a$ and $-a$ are square roots of a^2.

- If $a \geq 0$, then $+a$ is the nonnegative square root of a^2
- If $a < 0$, then $-a$ is the nonnegative square root of a^2

Since $\sqrt{a^2}$ denotes the nonnegative square root:

$$\sqrt{a^2} = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} = |a|$$

2.3 Properties of Absolute Value

Properties

If a and b are real numbers:

(a) $|-a| = |a|$

(b) $|ab| = |a||b|$

(c) $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

Proof (b):

$$|ab| = \sqrt{(ab)^2} = \sqrt{a^2 b^2} = \sqrt{a^2}\sqrt{b^2} = |a||b|$$

This extends to: $|a_1 a_2 \cdots a_n| = |a_1||a_2| \cdots |a_n|$ and $|a^n| = |a|^n$

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2.4 Geometric Interpretation

On a coordinate line, the distance d between points with coordinates a and b is:

$$d = |b - a|$$

Expression	Geometric Interpretation
$\|x - a\|$	Distance between x and a
$\|x + a\|$	Distance between x and $-a$
$\|x\|$	Distance between x and the origin

2.5 Inequalities of the Form absolute value of x minus a less than k and greater than k

Theorem

Inequality	Equivalent Form
$\|x - a\| < k$	$-k < x - a < k$ (i.e., $a - k < x < a + k$)
$\|x - a\| > k$	$x - a < -k$ or $x - a > k$ (i.e., $x < a - k$ or $x > a + k$)

Example 1: Solve absolute value of x minus 3 less than 4

$$-4 < x - 3 < 4$$
$$-1 < x < 7$$

Interval: $(-1, 7)$

Example 2: Solve absolute value of x plus 4 greater than or equal to 2

$$x + 4 \leq -2 \quad \text{or} \quad x + 4 \geq 2$$
$$x \leq -6 \quad \text{or} \quad x \geq -2$$

Set notation: $(-\infty, -6] \cup [-2, +\infty)$

2.6 The Triangle Inequality

Theorem 1.2.5 (Triangle Inequality)

If a and b are any real numbers, then:

$$|a + b| \leq |a| + |b|$$

Proof: Remember: $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$

Adding:

$$-(|a| + |b|) \leq a + b \leq |a| + |b|$$

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Case 1: If $a+b \geq 0$, then $|a+b| = a+b \leq |a|+|b|$

Case 2: If $a+b < 0$, then $|a+b| = -(a+b) \leq |a|+|b|$

Significance

This Triangle Inequality is the essence of the famous **Heisenberg Uncertainty Principle** in Quantum Physics.

3 Lesson 03: Coordinate Planes and Graphs

3.1 The Coordinate Plane

A **plane** is the intersection of two coordinate lines at 90 degrees. It is called the **coordinate plane**.

- The first letter in the name of the plane refers to the **horizontal axis**
- The second refers to the **vertical axis**

Quadrants

The coordinate axes divide the plane into four regions called **quadrants**, numbered counterclockwise:

- **Quadrant I:** $x > 0$, $y > 0$
- **Quadrant II:** $x < 0$, $y > 0$
- **Quadrant III:** $x < 0$, $y < 0$
- **Quadrant IV:** $x > 0$, $y < 0$

3.2 Graphs of Equations

Graph of an Equation

The **graph** of an equation in two variables x and y is the set of all points in the xy-plane whose coordinates are members of the solution set of the equation.

Is point 3 comma 2 a solution of 6x minus 4y equals 10?

$$6(3) - 4(2) = 18 - 8 = 10 \quad \checkmark$$

Yes, the point is a solution.

3.3 Intercepts

- Intersections with the x-axis: $(a, 0)$ — a is called an **x-intercept**
- Intersections with the y-axis: $(0, b)$ — b is called a **y-intercept**

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Find all intercepts of 3x plus 2y equals 6

x-intercept (set $y = 0$): $3x = 6 \Rightarrow x = 2 \Rightarrow (2, 0)$
y-intercept (set $x = 0$): $2y = 6 \Rightarrow y = 3 \Rightarrow (0, 3)$

3.4 Symmetry

Type	Test
Symmetric about x-axis	(x, y) and $(x, -y)$
Symmetric about y-axis	(x, y) and $(-x, y)$
Symmetric about origin	(x, y) and $(-x, -y)$

Example: y equals one eighth x to the fourth minus x squared

Substituting $-x$ for x:

$$y = \frac{1}{8}(-x)^4 - (-x)^2 = \frac{1}{8}x^4 - x^2$$

Equation unchanged $\Rightarrow$ **Symmetric about y-axis**

Example: x equals y squared

Substituting $-y$ for y:

$$x = (-y)^2 = y^2$$

Equation unchanged $\Rightarrow$ **Symmetric about x-axis**

4 Lesson 04: Lines

4.1 Slope

Definition 1.4.1 (Slope)

If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are points on a non-vertical line, then the slope m is:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Observations

- The slope of a **vertical line** is **UNDEFINED** (division by zero)
- It does not matter which point is P_1 and which is P_2
- Any two distinct points on a non-vertical line give the same slope

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Find the slope of the line through given points

(a) $(6, 2)$ and $(9, 8)$:

$$m = \frac{8-2}{9-6} = \frac{6}{3} = 2$$

(b) $(2, 9)$ and $(4, 3)$:

$$m = \frac{3-9}{4-2} = \frac{-6}{2} = -3$$

(c) $(-2, 7)$ and $(5, 7)$:

$$m = \frac{7-7}{5-(-2)} = \frac{0}{7} = 0$$

4.2 Angle of Inclination

Definition 1.4.2

For a line L not parallel to the x-axis, the **angle of inclination** is the smallest angle ϕ measured counterclockwise from the positive x-axis to L. For a line parallel to the x-axis, $\phi = 0$.

Theorem 1.4.3

For a non-vertical line:

$$m = \tan\phi$$

If the line is parallel to the y-axis, then $\phi = \pi/2$, so $\tan\phi$ is undefined.

Find angle of inclination

- If $m = 1$, then $\tan\phi = 1$, so $\phi = \pi/4$ (or $45°$)
- If $m = -1$, then $\tan\phi = -1$. Since $0 \leq \phi \leq \pi$, we get $\phi = 3\pi/4$ (or $135°$)

4.3 Parallel and Perpendicular Lines

Theorem 1.4.4

Let L_1 and L_2 be non-vertical lines with slopes m_1 and m_2:

(a) The lines are **parallel** if and only if $m_1 = m_2$

(b) The lines are **perpendicular** if and only if $m_1 m_2 = -1$

Perpendicular Slopes

Two non-vertical lines are perpendicular if and only if their slopes are **NEGATIVE RECIPROCALS** of one another:

$$m_2 = -\frac{1}{m_1}$$

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Show points A 1 comma 3

$$m_1 = \frac{7-3}{3-1} = \frac{4}{2} = 2 \quad \text{(slope through } A \text{ and } B\text{)}$$
$$m_2 = \frac{5-7}{7-3} = \frac{-2}{4} = -\frac{1}{2} \quad \text{(slope through } B \text{ and } C\text{)}$$
$$m_1 m_2 = (2)\left(-\frac{1}{2}\right) = -1$$

$\Rightarrow AB \perp BC$; thus ABC is a right triangle.

4.4 Equations of Lines

4.4.1 Lines Parallel to the Coordinate Axes

Theorem 1.4.5

The vertical line through $(a, 0)$: $x = a$
The horizontal line through $(0, b)$: $y = b$

4.4.2 Point-Slope Form

Theorem 1.4.6 (Point-Slope Form)

The line passing through $P_1(x_1, y_1)$ with slope m:

$$y - y_1 = m(x - x_1)$$

Line through point 2 comma 3 with slope negative 3 over 2

$$y - 3 = -\frac{3}{2}(x - 2)$$
$$y = -\frac{3}{2}x + 6$$

4.4.3 Slope-Intercept Form

Theorem 1.4.7 (Slope-Intercept Form)

The line with y-intercept b and slope m:

$$y = mx + b$$

Line with slope negative 9 crossing y-axis at point 0 comma negative 4

$$y = -9x - 4$$

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4.4.4 General Equation of a Line

Theorem 1.4.8

Every first-degree equation in x and y has a straight line as its graph and, conversely, every straight line can be represented by a first-degree equation:

$$Ax + By + C = 0$$

where A and B are not both zero.

Find slope and y-intercept of 8x plus 5y equals 20

$$5y = -8x + 20$$
$$y = -\frac{8}{5}x + 4$$

Slope $m = -\frac{8}{5}$, y-intercept $b = 4$

4.5 Applications

Fahrenheit-Celsius Conversion

Fahrenheit temperature (F) and Celsius temperature (C) are related by:

$$F = \frac{9}{5}C + 32$$

5 Lesson 05: Distance; Circles, Quadratic Equations

5.1 Distance Formula

Theorem 1.5.1 (Distance Formula)

The distance d between two points (x_1, y_1) and (x_2, y_2):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Find distance between point negative 2 comma 3 and point 1 comma 7

$$d = \sqrt{(1 - (-2))^2 + (7 - 3)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

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Show points A 4 comma 6

$$d(A,B) = \sqrt{(1-4)^2 + (-3-6)^2} = \sqrt{9+81} = \sqrt{90}$$
$$d(A,C) = \sqrt{(7-4)^2 + (5-6)^2} = \sqrt{9+1} = \sqrt{10}$$
$$d(B,C) = \sqrt{(7-1)^2 + (5-(-3))^2} = \sqrt{36+64} = 10$$

Since $[d(A,B)]^2 + [d(A,C)]^2 = 90 + 10 = 100 = [d(B,C)]^2$, ABC is a right triangle.

5.2 The Midpoint Formula

Theorem 1.5.2 (Midpoint Formula)

The midpoint of the line segment joining (x_1, y_1) and (x_2, y_2):

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Midpoint of point 3 comma negative 4 and point 7 comma 2

$$\text{Midpoint} = \left(\frac{3+7}{2}, \frac{-4+2}{2}\right) = \left(\frac{10}{2}, \frac{-2}{2}\right) = (5, -1)$$

5.3 Circles

Standard Form of a Circle

The circle of radius r centered at (x_0, y_0):

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

Circle of radius 4 centered at point negative 5 comma 3

$$(x-(-5))^2 + (y-3)^2 = 4^2$$
$$(x+5)^2 + (y-3)^2 = 16$$
$$x^2 + y^2 + 10x - 6y + 18 = 0$$

Circle with center point 1 comma negative 2 passing through point 4 comma 2

$$r = \sqrt{(1-4)^2 + (-2-2)^2} = \sqrt{9+16} = 5$$
$$(x-1)^2 + (y+2)^2 = 25$$
$$x^2 + y^2 - 2x + 4y - 20 = 0$$

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5.3.1 Finding Center and Radius

Equation	Center	Radius
$(x-2)^2+(y-5)^2=9$	$(2,5)$	3
$(x+7)^2+(y+1)^2=16$	$(-7,-1)$	4
$x^2+y^2=25$	$(0,0)$	5
$(x-4)^2+y^2=5$	$(4,0)$	$\sqrt{5}$

Unit Circle

The circle $x^2+y^2=1$, centered at the origin with radius 1, is called the **unit circle**.

Find center and radius of x squared plus y squared minus 8x plus 2y plus 8 equals 0

$$(x^2-8x)+(y^2+2y)=-8$$
$$(x^2-8x+16)+(y^2+2y+1)=-8+16+1$$
$$(x-4)^2+(y+1)^2=9$$

Center: $(4,-1)$, Radius: 3

5.3.2 Degenerate Cases of a Circle

Value of k	Graph
$k>0$	Circle with center (x_0,y_0) and radius $\sqrt{k}$
$k=0$	Single point (x_0,y_0)
$k<0$	No real graph

5.4 Quadratic Equations

Quadratic Equation

An equation of the form:

$$y=ax^2+bx+c \quad (a\neq 0)$$

is called a **quadratic equation** in x. The graph is a **parabola**.

- If $a>0$, the parabola opens **upward**
- If $a<0$, the parabola opens **downward**

Vertex Formula

The x-coordinate of the vertex:

$$x=-\frac{b}{2a}$$

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Sketch y equals negative x squared plus 4x minus 5

$a = -1$, $b = 4$, $c = -5$

$$x = -\frac{4}{2(-1)} = 2$$

Ball thrown upward: s equals 24.5t minus 4.9t squared

$$\text{Vertex: } t = -\frac{24.5}{2(-4.9)} = 2.5 \text{ sec}$$
$$s = 24.5(2.5) - 4.9(2.5)^2 = 30.625 \text{ m}$$

The ball rises **30.625 m** above the ground.

6 Lesson 06: Functions and Limits

6.1 Functions

Function

If a quantity y depends on another quantity x in such a way that **each value of x determines exactly one value of y**, we say that y is a function of x.

y equals 4x plus 1 is a function

Each value assigned to x determines a unique value of y.

x	$y = 4x + 1$
0	1
1	5
2	9
3	13

NOT A Function

$y = \pm\sqrt{x}$
If $x = 4$, then $y = \pm 2$. So a single value of x does not lead to exactly one value of y.

6.2 Notation for Functions

In the 1700's, Swiss mathematician **Euler** introduced the notation $y = f(x)$.

- $f(x)$ is read as "y function of x" NOT as "f multiplied by x"
- f does not represent a number; it expresses a functional relationship
- The variable alongside f is the **independent variable**
- The other is the **dependent variable**

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f of x equals x squared

$$f(3) = 9, \quad f(-2) = 4$$

phi of x equals 1 over x cubed minus 1

$$\phi(5^{1/6}) = \frac{1}{(5^{1/6})^3 - 1} = \frac{1}{5^{1/2} - 1} = \frac{1}{\sqrt{5} - 1}$$

$$\phi(1) = \frac{1}{1^3 - 1} = \frac{1}{0} = \text{undefined}$$

F of x equals 2x squared minus 1

$$F(d) = 2d^2 - 1$$
$$F(t-1) = 2(t-1)^2 - 1 = 2t^2 - 4t + 1$$

Remark

If two functions look alike in all aspects other than a difference in variables, then they are the **SAME** function. For example, $g(c) = c^2 - 4c$ and $g(x) = x^2 - 4x$ are the same.

6.3 Domain of a Function

Domain

The set consisting of all **allowable values** for the independent variable is called the **domain** of the function.

Cardboard problem

A square of side x cm is cut from four corners of a 10 cm square cardboard. The remaining area is:

$$y = 100 - 4x^2$$

Since x denotes length: $0 \leq x \leq 5$. So domain is $[0, 5]$.

6.3.1 Types of Domain

1. **Natural Domain:** Comes from the formula itself.
2. **Restricted Domain:** Altered by algebraic manipulation.

Natural Domain: h of x equals 1 over (x minus 1)(x minus 3)

$x = 1$ and $x = 3$ are not in the domain.

$$\text{Domain: } (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

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Restricted Domain: h of x equals (x squared minus 4) over (x minus 2)

Original: $x \neq 2$. After simplification: $h(x) = x + 2$. Now $h(2) = 4$ is defined.

$$h(x) = x + 2, \quad x \neq 2$$

6.4 Range of a Function

Range

The set of all possible values for $f(x)$ as x varies over the domain is called the **range**.

Range of f of x equals x squared

As x varies over reals, y is all non-negative reals. Range: $[0, +\infty)$.

Range of g of x equals 2 plus square root of x minus 1

Domain: $[1, +\infty)$. As x varies, $\sqrt{x-1}$ varies over $[0, +\infty)$, so $y = 2 + \sqrt{x-1}$ varies over $[2, +\infty)$.

Range of y equals (x plus 1) over (x minus 1)

Solving for x:

$$\begin{aligned} y(x-1) &= x+1 \\ yx - y &= x+1 \\ x(y-1) &= 1+y \\ x &= \frac{y+1}{y-1} \end{aligned}$$

It is evident that $y = 1$ is not in the range:

$$\text{Range: } (-\infty, 1) \cup (1, +\infty)$$

6.5 Functions Defined Piecewise

Taxicab Ride Cost

$$f(x) = \begin{cases} 1.75 & 0 < x \leq 1 \\ 1.75 + 0.50(x-1) & x > 1 \end{cases}$$

6.6 Reversing the Roles of x and y

3x plus 2y equals 6

$$y = -\frac{3}{2}x + 3 \quad \text{or} \quad x = -\frac{2}{3}y + 2$$

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7 Lesson 07: Operations on Functions

7.1 Arithmetic Operations on Functions

Operations

Operation	Definition
Sum	$(f+g)(x) = f(x) + g(x)$
Difference	$(f-g)(x) = f(x) - g(x)$
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$(f/g)(x) = f(x)/g(x)$

Domain Notes

- For $f+g$, $f-g$, $f \cdot g$: domain is the **intersection** of domains of f and g
- For f/g: intersection of domains, **except** where $g(x) = 0$

f of x equals 1 plus square root of x minus 2

$$(f+g)(x) = x + \sqrt{x-2}$$

Domain of f: $[2, +\infty)$; Domain of g: $(-\infty, +\infty)$
Domain of $f+g$: $[2, +\infty)$

f of x equals 3 square root of x

$$(f \cdot g)(x) = 3x, \quad x \geq 0$$

7.2 Notation

$$f^2(x) = f(x) \cdot f(x), \quad f^n(x) = \underbrace{f(x) \cdot f(x) \cdots f(x)}_{n \text{ times}}$$

7.3 Composition of Functions

Composition

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ consists of all x in the domain of g for which $g(x)$ is in the domain of f.

f of x equals x squared plus 3

$$(f \circ g)(x) = (\sqrt{x})^2 + 3 = x + 3$$

Domain: $[0, +\infty)$

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Important

$$(f \circ g) \neq (g \circ f) \quad \text{generally}$$

"Put Sock on then Shoe ≠ Put Shoe on first then Sock."

7.4 Expressing Functions as a Decomposition

h of x equals (x plus 1) squared

$$f(x) = x + 1$$
$$g(x) = x^2$$
$$h(x) = g(f(x))$$

Function	Inside $g(x)$	Outside $f(x)$	Composition
$\sin^2 x$	$\sin x$	x^2	$f(g(x))$
$\sin(x^3)$	x^3	$\sin x$	$f(g(x))$
$\tan^3 x$	$\tan x$	x^3	$f(g(x))$

7.5 Classification of Functions

Constant Functions

These assign the same **number** to every x in the domain.
Example: $f(x) = 2$, so $f(1) = 2$, $f(-7) = 2$, etc.

Monomial in x

Anything that looks like cx^n with c a constant and n a **non-negative integer**.
Examples: $2x^5$, $3x^{55}$
NOT monomials: $4x^{-4}$, $5x^{1/2}$

Polynomial in x

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Examples: $4x^4 + 3x^2 + 1$, $17 - \frac{4}{3}x^2$

8 Lesson 08: Graphing Functions

Graph of a Function

The graph of a function f in the xy-plane is the graph of the equation $y = f(x)$.

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Sketch f of x equals absolute value of x

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

The absolute value function is **PIECEWISE** defined.

t of x equals (x squared minus 4) over (x minus 2)

$$t(x) = x + 2, \quad x \neq 2$$

There is a **HOLE** in the graph at $x = 2$.

8.1 Graphing Functions by Translations

Translations

Transformation	Geometric Effect
$y = f(x) + c$	Translation **UP** by c units
$y = f(x) - c$	Translation **DOWN** by c units
$y = f(x + c)$	Translation **LEFT** by c units
$y = f(x - c)$	Translation **RIGHT** by c units

Sketch y equals square root of x minus 3 plus 2

1. Translate $y = \sqrt{x}$ 3 units RIGHT: $y = \sqrt{x - 3}$
2. Translate this 2 units UP: $y = \sqrt{x - 3} + 2$

Sketch y equals x squared minus 4x plus 5

Complete the square:

$$y = (x^2 - 4x + 4) + 5 - 4 = (x - 2)^2 + 1$$

Shift $y = x^2$ RIGHT 2 units, then UP 1 unit.

8.2 Reflections

Reflections

- Graphs of $y = f(x)$ and $y = f(-x)$ are reflections about the y-**axis**
- Graphs of $y = f(x)$ and $y = -f(x)$ are reflections about the x-**axis**

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8.3 Vertical Scaling

Vertical Scaling

If $f(x)$ is multiplied by a positive constant c:

- Graph is **COMPRESSED** vertically if $0 < c < 1$
- Graph is **STRETCHED** vertically if $c > 1$

8.4 Vertical Line Test

Vertical Line Test

A graph in the plane is the graph of a function **if and only if** NO vertical line intersects the graph more than once.

x squared plus y squared equals 25

The graph is a **circle**. Vertical lines cross in more than 2 places. So it is NOT a function of x.

9 Lesson 09: Limits (Intuitive Introduction)

Two Branches of Calculus

- Calculus from the **tangent problem** $\rightarrow$ **DIFFERENTIAL CALCULUS**
- Calculus from the **area problem** $\rightarrow$ **INTEGRAL CALCULUS**

Both depend on the fundamental notion of **LIMIT**.

9.1 Tangent Lines and Limits

Consider a point P on a curve. Let Q be another point on the curve. Draw a line through P and Q — this is the **SECANT** line. Now move Q toward P. The secant line rotates to a "limiting" position. The line in this limiting position is called the **TANGENT** line at P.

9.2 Limits

Limit

Limits are a way to study the behavior of the y-values of a function in response to the x-values as they approach some number or go to infinity.

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Limit as x approaches 0 of sine x over x

$f(x) = \frac{\sin(x)}{x}$ is not defined at $x = 0$.

$$\lim_{x \to 0^+} \frac{\sin(x)}{x} = 1 \quad \text{(Right Hand Limit)}$$

$$\lim_{x \to 0^-} \frac{\sin(x)}{x} = 1 \quad \text{(Left Hand Limit)}$$

When both match: $\lim_{x \to 0} \frac{\sin(x)}{x} = 1$

9.3 Table of Limit Notations

Notation	Meaning
$\lim_{x \to a^+} f(x) = L$	Limit as x approaches a from the right is L
$\lim_{x \to a^-} f(x) = L$	Limit as x approaches a from the left is L
$\lim_{x \to a} f(x) = L$	Limit as x approaches a (both sides)
$\lim_{x \to +\infty} f(x) = L$	Limit as $x \to +\infty$ is L
$\lim_{x \to -\infty} f(x) = L$	Limit as $x \to -\infty$ is L

9.4 Existence of Limits

Limit Does Not Exist (DNE)

Limits fail for many reasons:

1. **Oscillations**
2. **Unbounded increase or decrease**

Unbounded increase

$$\lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x) = \lim_{x \to x_0} f(x) = +\infty$$

The limit **FAILS to EXIST** because of unboundedness. The $+\infty$ is NOT a number!

10 Lesson 10: Limits (Computational Techniques)

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10.1 Basic Limits

Limit	Value
$\lim_{x\to a} k$	k
$\lim_{x\to a^+} k$	k
$\lim_{x\to a^-} k$	k
$\lim_{x\to +\infty} k$	k
$\lim_{x\to a} x$	a
$\lim_{x\to +\infty} x$	$+\infty$
$\lim_{x\to -\infty} x$	$-\infty$

Theorem 2.5.1

If $L_1 = \lim f(x)$ and $L_2 = \lim g(x)$ both exist, then:

$$\lim[f(x) + g(x)] = L_1 + L_2$$
$$\lim[f(x) - g(x)] = L_1 - L_2$$
$$\lim[f(x) \cdot g(x)] = L_1 \cdot L_2$$
$$\lim \frac{f(x)}{g(x)} = \frac{L_1}{L_2} \quad (L_2 \neq 0)$$

Key Results

$$\lim_{x\to a}(x^n) = a^n$$
$$\lim[k \cdot g(x)] = k \cdot \lim g(x)$$

A constant factor can be moved through a limit sign.

Limit as x approaches 5 of x squared minus 4x plus 3

$$= \lim_{x\to 5} x^2 - 4 \lim_{x\to 5} x + \lim_{x\to 5} 3$$
$$= (5)^2 - 4(5) + 3 = 25 - 20 + 3 = 8$$

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10.2 Limits Involving 1 over x

Limit	Value
$\lim_{x\to 0^+} \frac{1}{x}$	$+\infty$
$\lim_{x\to 0^-} \frac{1}{x}$	$-\infty$
$\lim_{x\to +\infty} \frac{1}{x}$	0
$\lim_{x\to -\infty} \frac{1}{x}$	0

10.3 Limits of Rational Functions as x approaches a

Limit as x approaches 2 of (5x cubed plus 4) over (x minus 3)

$$= \frac{5(2)^3 + 4}{2 - 3} = \frac{44}{-1} = -44$$

Limit as x approaches 2 of (x squared minus 4) over (x minus 2)

$$= \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2}(x+2) = 4$$

10.4 Quick Rule for Limits as x approaches plus or minus infinity

Quick Rule

$$\lim_{x\to\pm\infty} \frac{c_0 + c_1 x + \cdots + c_n x^n}{d_0 + d_1 x + \cdots + d_m x^m} = \lim_{x\to\pm\infty} \frac{c_n x^n}{d_m x^m}$$

Not true if $x \to a$ (finite number).

Limit as x approaches negative infinity of (4x squared minus x) over (2x cubed minus 5)

$$= \lim_{x\to -\infty} \frac{4x^2}{2x^3} = \lim_{x\to -\infty} \frac{2}{x} = 0$$

11 Lesson 11: Limits (Rigorous Approach)

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11.1 Epsilon-Delta Definition

Formal Definition of Limit

For any number $\epsilon > 0$, if we can find an open interval (x_0, x_1) on the x-axis containing a point a such that

$$L - \epsilon < f(x) < L + \epsilon$$

for each x in (x_0, x_1) except possibly $x = a$, then we say

$$\lim_{x \to a} f(x) = L$$

Equivalent Forms

$$|f(x) - L| < \epsilon$$
$$0 < |x - a| < \delta$$

Find delta such that limit as x approaches 2 of (3x minus 5) equals 1

$$\begin{aligned} |(3x - 5) - 1| &< \epsilon \\ |3x - 6| &< \epsilon \\ 3|x - 2| &< \epsilon \\ |x - 2| &< \frac{\epsilon}{3} \end{aligned}$$

Let $\delta = \epsilon/3$. Then $|x - 2| < \epsilon/3$ if $0 < |x - 2| < \epsilon/3$.

Show limit as x approaches 0 of f of x does not exist

Suppose the limit exists and equals L. Take $\epsilon = 1$. There is $\delta > 0$ such that:

$$|f(x) - L| < 1 \quad \text{if } 0 < |x| < \delta$$

But $x = \delta/2$ and $x = -\delta/2$ both satisfy this, so:

$$|1 - L| < 1 \quad \text{and} \quad |-1 - L| < 1$$

$$0 < L < 2 \quad \text{and} \quad -2 < L < 0$$

Contradiction! So the limit does not exist.

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12 Lesson 12: Continuity

Definition 2.7.1 (Continuity)

A function f is **continuous** at $x = c$ if:

(a) $f(c)$ is defined

(b) $\lim_{x \to c} f(x)$ exists

(c) $\lim_{x \to c} f(x) = f(c)$

If any condition fails, f is **discontinuous** at c.

f of x equals (x squared minus 4) over (x minus 2)

Discontinuous at $x = 2$ because $f(2)$ is undefined.

g of x piecewise: (x squared minus 4)/(x minus 2) if x not equal 2

$$\lim_{x \to 2} g(x) = \lim_{x \to 2} (x + 2) = 4 \neq g(2) = 3$$

So g is discontinuous at $x = 2$.

Theorem 2.7.2

Polynomials are continuous functions.

Theorem 2.7.3 (Properties)

If f and g are continuous at c, then:

(a) $f + g$ is continuous at c

(b) $f - g$ is continuous at c

(c) $f \cdot g$ is continuous at c

(d) f/g is continuous at c if $g(c) \neq 0$

Theorem 2.7.5 (Composition)

If $\lim g(x) = L$ and f is continuous at L, then:

$$\lim f(g(x)) = f(L)$$

That is:

$$\lim f(g(x)) = f(\lim g(x))$$

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Definition 2.7.7 (One-Sided Continuity)

Continuous from the left at c:

1. $f(c)$ is defined
2. $\lim_{x \to c^-} f(x)$ exists
3. $\lim_{x \to c^-} f(x) = f(c)$

Continuous from the right at c:

1. $f(c)$ is defined
2. $\lim_{x \to c^+} f(x)$ exists
3. $\lim_{x \to c^+} f(x) = f(c)$

Definition 2.7.8 (Continuity on closed interval a to b)

f is continuous on $[a, b]$ if:

1. f is continuous on (a, b)
2. f is continuous from the right at a
3. f is continuous from the left at b

Theorem 2.7.9 (Intermediate Value Theorem)

If f is continuous on $[a, b]$ and C is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number x in $[a, b]$ such that $f(x) = C$.

Theorem 2.7.10

If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there is at least one solution of $f(x) = 0$ in (a, b).

13 Lesson 13: Limits and Continuity of Trigonometric Functions

Theorem 2.8.1

The functions $\sin(x)$ and $\cos(x)$ are continuous.

$$\lim_{x \to 0} \sin(x) = 0, \quad \lim_{x \to 0} \cos(x) = 1$$

Proof: Let $h = x - c$. Then $x \to c$ is equivalent to $h \to 0$.

$$\begin{aligned} \lim_{h \to 0} \sin(c + h) &= \lim_{h \to 0} [\sin(c)\cos(h) + \cos(c)\sin(h)] \\ &= \sin(c) \lim_{h \to 0} \cos(h) + \cos(c) \lim_{h \to 0} \sin(h) \\ &= \sin(c)(1) + \cos(c)(0) = \sin(c) \end{aligned}$$

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13.1 Continuity of Other Trigonometric Functions

Continuity of Trig Functions

- $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$ — continuous except where $\cos(x) = 0$
- $\cot(x) = \dfrac{\cos(x)}{\sin(x)}$ — continuous except where $\sin(x) = 0$
- $\sec(x) = \dfrac{1}{\cos(x)}$ — continuous except where $\cos(x) = 0$
- $\csc(x) = \dfrac{1}{\sin(x)}$ — continuous except where $\sin(x) = 0$

13.2 Squeeze Theorem

Squeeze Theorem

Let f, g, and h be functions satisfying:

$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing a, with possible exception at a. If

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} f(x) = L$$

Limit as x approaches 0 of x squared sine squared 1 over x

Since $0 \leq \sin^2(1/x) \leq 1$:

$$0 \leq x^2 \sin^2(1/x) \leq x^2$$

$$\lim_{x \to 0} 0 = \lim_{x \to 0} x^2 = 0$$

So by Squeeze Theorem:

$$\lim_{x \to 0} x^2 \sin^2(1/x) = 0$$

13.3 Important Trigonometric Limits

Theorem 2.8.3

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

$$\lim_{x \to 0} \frac{1 - \cos(x)}{x} = 0$$

Proof of first limit:

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Construct angle x in standard position in a unit circle. From the figure:

$$0 < \text{Area of } \triangle OBP < \text{Area of sector } OBP < \text{Area of } \triangle OBQ$$

$$\frac{1}{2}\sin(x) < \frac{1}{2}x < \frac{1}{2}\tan(x)$$

Multiplying by 2 and dividing by $\sin(x)$:

$$1 < \frac{x}{\sin(x)} < \frac{1}{\cos(x)}$$

Taking reciprocals:

$$\cos(x) < \frac{\sin(x)}{x} < 1$$

As $x \to 0$: $\cos(x) \to 1$ and $1 \to 1$. By Squeeze Theorem:

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$

Limits of sine and cosine as x approaches plus or minus infinity

The y-values oscillate between 1 and -1 as $x \to \pm\infty$, so the limits **DNE**.

14 Lesson 14: Tangent Lines, Rates of Change

14.1 Slope of Secant Line

If $P(x_0, y_0)$ and $Q(x_1, y_1)$ are distinct points on a curve $y = f(x)$:

$$m_{\text{sec}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

14.2 Slope of Tangent Line

As $x_1 \to x_0$, Q approaches P, and the secant line approaches the tangent line:

$$m_{\tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

14.3 Average Velocity

$$\text{Average Velocity} = \frac{\text{distance traveled}}{\text{Time Elapsed}} = \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

Instantaneous Velocity

$$v_{\text{inst}} = \lim_{t_1 \to t_0} v_{\text{ave}} = \lim_{t_1 \to t_0} \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

This is the slope of the tangent line at (t_0, d_0).

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14.4 Definitions

Definition 3.1.1 (Average Rate of Change)

If $y = f(x)$, the **average rate of change** of y w.r.t x over $[x_0, x_1]$:

$$m_{\text{sec}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Definition 3.1.1 (Instantaneous Rate of Change)

The **instantaneous rate of change** of y w.r.t x at x_0:

$$m_{\tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

y equals x squared plus 1

(a) Average rate over interval 3 to 5:

$$m_{\text{sec}} = \frac{f(5) - f(3)}{5 - 3} = \frac{26 - 10}{2} = 8$$

(b) Instantaneous rate at x naught equals negative 4:

$$\begin{aligned} m_{\tan} &= \lim_{x_1 \to -4} \frac{(x_1^2 + 1) - 17}{x_1 + 4} \\ &= \lim_{x_1 \to -4} \frac{x_1^2 - 16}{x_1 + 4} \\ &= \lim_{x_1 \to -4} (x_1 - 4) = -8 \end{aligned}$$

(c) Instantaneous rate at general point:

$$m_{\tan} = \lim_{x_1 \to x_0} \frac{x_1^2 - x_0^2}{x_1 - x_0} = \lim_{x_1 \to x_0} (x_1 + x_0) = 2x_0$$

15 Lesson 15: The Derivative

Definition 3.2.1 (Tangent Line)

If $P(x_0, y_0)$ is a point on the graph of f, the tangent line at P has slope:

$$m_{\tan} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Equation of tangent line: $y - y_0 = m_{\tan}(x - x_0)$

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Find slope and equation of tangent line to x squared at point 3 comma 9

$$\begin{aligned} m_{\tan} &= \lim_{h\to 0} \frac{(3+h)^2 - 9}{h} \\ &= \lim_{h\to 0} \frac{9 + 6h + h^2 - 9}{h} \\ &= \lim_{h\to 0} \frac{h(6+h)}{h} = \lim_{h\to 0}(6+h) = 6 \end{aligned}$$

Equation: $y - 9 = 6(x-3) \Rightarrow y = 6x - 9$

Definition 3.2.2 (Derivative)

The **derivative** of f with respect to x:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

The domain of f' consists of all x for which the limit exists.

Two Interpretations of Derivative

1. **Geometric:** f' is the function whose value at x is the slope of the tangent line to the graph of f at x
2. **Rate of Change:** If $y = f(x)$, then f' is the instantaneous rate of change of y w.r.t x

f of x equals x squared plus 1

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{[(x+h)^2 + 1] - [x^2 + 1]}{h} \\ &= \lim_{h\to 0} \frac{2xh + h^2}{h} = \lim_{h\to 0}(2x + h) = 2x \end{aligned}$$

$f'(2) = 4$, $f'(0) = 0$, $f'(-2) = -4$

f of x equals square root of x

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h\to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \end{aligned}$$

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15.1 Derivative Notation

Notation	Meaning
$f'(x)$	Derivative of f w.r.t x
$\frac{d}{dx}[f(x)]$	Derivative of f w.r.t x
$\frac{dy}{dx}$	Derivative of y w.r.t x

15.2 Existence of Derivatives

Non-Differentiability

Non-differentiability occurs when the graph of $f(x)$ has:

1. **Corners**
2. **Vertical tangents**
3. **Points of discontinuity**

Theorem 3.2.3

If f is differentiable at x_0, then f is also **continuous** at x_0.

Proof:

$$\lim_{h\to 0}[f(x_0+h)-f(x_0)] = \lim_{h\to 0}\left[\frac{f(x_0+h)-f(x_0)}{h}\cdot h\right]$$
$$= f'(x_0)\cdot 0 = 0$$

So $\lim_{h\to 0} f(x_0+h) = f(x_0)$.

f of x equals absolute value of x

$$f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

At $x = 0$, derivative does not exist because left and right limits differ.

16 Lesson 16: Techniques of Differentiation

Theorem 3.3.1 (Constant Rule)

If $f(x) = c$ for all x, then:

$$\frac{d}{dx}[c] = 0$$

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Theorem 3.3.2 (Power Rule)

If n is a positive integer:

$$\frac{d}{dx}[x^n] = n \cdot x^{n-1}$$

Power Rule Examples

$$\frac{d}{dx}[x^5] = 5x^4, \quad \frac{d}{dx}[x] = 1$$

Theorem 3.3.3 (Constant Multiple Rule)

$$\frac{d}{dx}[cf(x)] = c \cdot \frac{d}{dx}[f(x)]$$

Derivative of 3x to the eighth

$$\frac{d}{dx}[3x^8] = 3 \cdot 8x^7 = 24x^7$$

Theorem 3.3.4 (Sum and Difference Rule)

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$$

Derivative of x to the fourth plus x cubed

$$\frac{d}{dx}[x^4 + x^3] = 4x^3 + 3x^2$$

Theorem 3.3.5 (Product Rule)

$$\frac{d}{dx}[f(x) \cdot g(x)] = f(x) \cdot \frac{d}{dx}[g(x)] + g(x) \cdot \frac{d}{dx}[f(x)]$$

In short: $(fg)' = fg' + gf'$

Theorem 3.3.6 (Quotient Rule)

If $g(x) \neq 0$:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x) \cdot \frac{d}{dx}[f(x)] - f(x) \cdot \frac{d}{dx}[g(x)]}{[g(x)]^2}$$

In short: $\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$

Theorem 3.3.7 (Reciprocal Rule)

If $g(x) \neq 0$:

$$\frac{d}{dx}\left[\frac{1}{g(x)}\right] = -\frac{\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

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Theorem 3.3.8 (Generalized Power Rule)

If n is any integer:

$$\frac{d}{dx}[x^n] = n \cdot x^{n-1}$$

17 Lesson 17: Derivatives of Trigonometric Functions

Derivative of sine x

$$\frac{d}{dx}[\sin(x)] = \cos(x)$$

Proof:

$$\begin{aligned}
\frac{d}{dx}[\sin(x)] &= \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h} \\
&= \lim_{h\to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h} \\
&= \sin(x) \lim_{h\to 0} \frac{\cos(h) - 1}{h} + \cos(x) \lim_{h\to 0} \frac{\sin(h)}{h} \\
&= \sin(x)(0) + \cos(x)(1) = \cos(x)
\end{aligned}$$

Derivative of cosine x

$$\frac{d}{dx}[\cos(x)] = -\sin(x)$$

Derivative of tangent x

$$\frac{d}{dx}[\tan(x)] = \sec^2(x)$$

Proof:

$$\begin{aligned}
\frac{d}{dx}[\tan(x)] &= \frac{d}{dx}\left[\frac{\sin(x)}{\cos(x)}\right] \\
&= \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)} \\
&= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x)
\end{aligned}$$

Derivative of secant x

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$$

Derivative of cosecant x

$$\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$$

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Derivative of cotangent x

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

Summary of Trigonometric Derivatives

Function	Derivative
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$
$\sec(x)$	$\sec(x)\tan(x)$
$\csc(x)$	$-\csc(x)\cot(x)$
$\cot(x)$	$-\csc^2(x)$

Related Rates with Trigonometry

A building 100 ft high. Sun's angle of elevation θ. Shadow length x:

$$\tan\theta = \frac{100}{x} \Rightarrow x = 100\cot\theta$$

$$\frac{dx}{d\theta} = -100\csc^2(\theta)$$

At $\theta = 45° = \pi/4$:

$$\frac{dx}{d\theta} = -100\csc^2(\pi/4) = -100(2) = -200 \text{ ft/radian}$$

Converting to degrees:

$$-200 \cdot \frac{\pi}{180} = -\frac{10\pi}{9} \text{ ft/degree}$$

18 Lesson 18: The Chain Rule

Theorem 3.5.2 (Chain Rule)

If g is differentiable at x and f is differentiable at $g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

where $y = f(u)$ and $u = g(x)$.

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Generalized Derivative Formula

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

In words: "Derivative of the OUTER function f, then Derivative of the INNER function g, and multiply the two together."

Derivative of 4 cosine x cubed

Let $u = x^3$:

$$= (-4\sin(u))(3x^2) = -12x^2\sin(x^3)$$

Derivative of (x squared minus x plus 1) to the 23rd power

Let $u = x^2 - x + 1$:

$$= 23u^{22}(2x-1) = 23(x^2 - x + 1)^{22}(2x-1)$$

Derivative of sine 2x

$$= \cos(2x) \cdot 2 = 2\cos(2x)$$

Derivative of tangent (x squared plus 1)

$$= \sec^2(x^2+1) \cdot 2x = 2x\sec^2(x^2+1)$$

19 Lesson 19: Implicit Differentiation

Implicit Differentiation

When y is not alone on one side, differentiate both sides w.r.t x, treating y as a function of x (using Chain Rule).

xy equals 1

Method 1 (Solve first):

$$y = \frac{1}{x} \Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$$

Method 2 (Implicit):

$$x\frac{dy}{dx} + y(1) = 0$$
$$x\frac{dy}{dx} = -y$$
$$\frac{dy}{dx} = -\frac{y}{x} = -\frac{1/x}{x} = -\frac{1}{x^2}$$

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5y squared plus sine y equals x squared

$$5(2y)\frac{dy}{dx} + \cos(y)\frac{dy}{dx} = 2x$$
$$\frac{dy}{dx}(10y + \cos y) = 2x$$
$$\frac{dy}{dx} = \frac{2x}{10y + \cos y}$$

7y to the fourth plus x cubed y plus x equals 4

$$28y^3\frac{dy}{dx} + x^3\frac{dy}{dx} + 3yx^2 + 1 = 0$$
$$\frac{dy}{dx}(28y^3 + x^3) = -3yx^2 - 1$$
$$\frac{dy}{dx} = -\frac{3yx^2 + 1}{28y^3 + x^3}$$

At $(4, 0)$: $m_{\tan} = -\frac{1}{64}$

4x squared minus 2y squared equals 9

$$8x - 4y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2x}{y}$$
$$\frac{d^2y}{dx^2} = \frac{y(2) - 2x(2x/y)}{y^2} = \frac{2y^2 - 4x^2}{y^3} = -\frac{9}{y^3}$$

Power Rule for Rational Powers

$$\frac{d}{dx}[x^r] = r \cdot x^{r-1}$$

where r is a rational number.

20 Lesson 20: Derivatives of Logarithmic and Exponential Functions

Logarithm

$y = \log_b(x)$ means $b^y = x$. b is the base.

Theorem 7.1.1

$$e = \lim_{x \to 0}(1 + x)^{1/x} \approx 2.71$$

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Derivative of Logarithmic Function

$$\frac{d}{dx}[\log_b(x)] = \frac{1}{x}\log_b(e) = \frac{1}{x\ln(b)}, \quad x > 0$$

Natural Log Derivative

$$\frac{d}{dx}[\ln(x)] = \frac{1}{x}, \quad x > 0$$

Generalized Formulas

$$\frac{d}{dx}[\log_b(u)] = \frac{1}{u\ln(b)} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}[\ln(u)] = \frac{1}{u} \cdot \frac{du}{dx}$$

Derivative of ln of (x squared plus 1)

$$= \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}$$

20.1 Logarithmic Differentiation

y equals x squared times cube root of (7x minus 14) over (1 plus x squared) to the fourth

$$\ln y = 2\ln x + \frac{1}{3}\ln(7x-14) - 4\ln(1+x^2)$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2}$$

$$\frac{dy}{dx} = \left(\frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2}\right) \cdot y$$

Power Rule for Any Real Number

$$\frac{d}{dx}[x^r] = r \cdot x^{r-1}$$

for any real number r.

Derivative of Exponential Functions

$$\frac{d}{dx}[b^x] = b^x \ln(b)$$

$$\frac{d}{dx}[e^x] = e^x$$

In general:

$$\frac{d}{dx}[e^u] = e^u \cdot \frac{du}{dx}$$

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20.2 Inverse Functions

Definition 7.4.1

Two functions f and g are inverses if:

$$f(g(x)) = x \quad \text{and} \quad g(f(x)) = x$$

Denoted as f^{-1}.

Theorem 7.4.7 (Derivative of Inverse)

$$\frac{dy}{dx} = \frac{1}{dx/dy}$$

21 Lesson 21: Applications of Differentiation

21.1 Related Rates

Steps for Solving Related Rate Problems

1. Draw a figure and label changing quantities
2. Identify known and unknown rates of change
3. Find an equation relating the quantities
4. Differentiate w.r.t the variable (usually time)
5. Evaluate at appropriate points

Oil Spill

Radius increases at 2 ft/sec. Find rate of area increase when $r = 60$ ft.

$$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left.\frac{dA}{dt}\right|_{r=60} = 2\pi(60)(2) = 240\pi \text{ ft}^2/\text{sec}$$

Ladder Problem

5 ft ladder, base moving at 2 ft/sec. Find rate top is moving down when base is 4 ft from wall.

$$x^2 + y^2 = 25$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$

When $x = 4$, $y = 3$:

$$\frac{dy}{dt} = -\frac{4}{3}(2) = -\frac{8}{3} \text{ ft/sec}$$

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21.2 Increasing and Decreasing Functions

Definition 4.2.1

(a) f is **increasing** if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$

(b) f is **decreasing** if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$

(c) f is **constant** if $f(x_1) = f(x_2)$ for all x_1, x_2

Theorem 4.2.1

Let f be continuous on $[a, b]$ and differentiable on (a, b):

(a) If $f'(x) > 0$ for all x in (a, b), then f is **increasing** on $[a, b]$

(b) If $f'(x) < 0$ for all x in (a, b), then f is **decreasing** on $[a, b]$

(c) If $f'(x) = 0$ for all x in (a, b), then f is **constant** on $[a, b]$

22 Lesson 22: Relative Extrema

Definition 4.3.1 (Relative Maximum)

f has a **relative maximum** at x_0 if $f(x_0) \geq f(x)$ for all x in some open interval containing x_0.

Definition 4.3.2 (Relative Minimum)

f has a **relative minimum** at x_0 if $f(x_0) \leq f(x)$ for all x in some open interval containing x_0.

Definition 4.3.3 (Relative Extremum)

f has a **relative extremum** at x_0 if it has either a relative maximum or relative minimum at x_0.

Definition 4.3.5 (Critical Point)

A number x_0 in the domain of f is a **critical point** if either $f'(x_0) = 0$ or f is not differentiable at x_0.

Theorem 4.3.4

If f has a relative extremum at x_0, then x_0 is a critical point of f.

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Theorem 4.3.6 (First Derivative Test)

Suppose f is continuous at a critical point x_0:

(a) If f' changes from $+$ to $-$ at x_0: **relative maximum**

(b) If f' changes from $-$ to $+$ at x_0: **relative minimum**

(c) If f' has same sign on both sides: **no relative extremum**

Theorem 4.3.7 (Second Derivative Test)

Suppose f is twice differentiable at x_0 and $f'(x_0) = 0$:

(a) If $f''(x_0) < 0$: **relative maximum**

(b) If $f''(x_0) > 0$: **relative minimum**

(c) If $f''(x_0) = 0$: **inconclusive**

f of x equals x to the fourth minus 2x squared

$$f'(x) = 4x^3 - 4x = 4x(x-1)(x+1)$$
$$f''(x) = 12x^2 - 4$$

Stationary points: $x = 0, \pm 1$

$$f''(0) = -4 < 0 \Rightarrow \text{relative max at } x = 0$$
$$f''(1) = 8 > 0 \Rightarrow \text{relative min at } x = 1$$
$$f''(-1) = 8 > 0 \Rightarrow \text{relative min at } x = -1$$

22.1 Graphs of Rational Functions

Rational Function

$$R(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials.

f of x equals (x squared plus 2x) over (x squared minus 1)

Vertical asymptotes: $x^2 - 1 = 0 \Rightarrow x = \pm 1$
Horizontal asymptote:

$$\lim_{x \to \pm\infty} \frac{x^2 + 2x}{x^2 - 1} = \lim_{x \to \pm\infty} \frac{x^2}{x^2} = 1$$

So $y = 1$ is the horizontal asymptote.

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23 Lesson 23: Maximum and Minimum Values of Functions

Absolute Extrema

- **Absolute maximum:** maximum value over the entire domain
- **Absolute minimum:** minimum value over the entire domain

Theorem 4.6.4 (Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f has both an absolute maximum and an absolute minimum on $[a, b]$.

Steps for Finding Absolute Extrema

1. Find critical points of f in (a, b)
2. Evaluate f at all critical points and endpoints a, b
3. The largest value is the maximum; the smallest is the minimum

f of x equals 2x cubed minus 15x squared plus 36x on interval 1 to 5

$$f'(x) = 6x^2 - 30x + 36 = 6(x-3)(x-2) = 0$$

Critical points: $x = 2, 3$

x	$f(x)$
1	23
2	28
3	27
5	55

Max is 55 at $x = 5$; min is 23 at $x = 1$.

Theorem 4.6.5

If f is continuous on $(-\infty, +\infty)$ and has an absolute extremum, then it must occur at a critical point of f.

f of x equals x to the fourth plus 2x cubed minus 1 on all reals

$$\lim_{x \to \pm\infty} (x^4 + 2x^3 - 1) = +\infty$$

So f has a minimum but no maximum.

$$f'(x) = 4x^3 + 6x^2 = 2x^2(2x+3) = 0$$

Critical points: $x = 0, -\frac{3}{2}$

$$f(0) = -1, \quad f\left(-\frac{3}{2}\right) = -\frac{43}{16}$$

Min $= -\frac{43}{16}$ at $x = -\frac{3}{2}$.

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23.1 Applied Maximum and Minimum Problems

Rectangle with perimeter 100 ft

Let x = length, y = width.

$$2x + 2y = 100 \Rightarrow y = 50 - x$$

$$A = xy = x(50 - x) = 50x - x^2$$

$$\frac{dA}{dx} = 50 - 2x = 0 \Rightarrow x = 25$$

$y = 25$ also. So the rectangle is a **square** with sides 25 ft.

Open box from 16 in by 30 in cardboard

Cut squares of side x from corners:

$$V = x(16 - 2x)(30 - 2x) = 4x^3 - 92x^2 + 480x$$

$$\frac{dV}{dx} = 12x^2 - 184x + 480 = 0$$

$$3x^2 - 46x + 120 = 0$$

$$x = \frac{46 \pm \sqrt{2116 - 1440}}{6} = \frac{46 \pm 26}{6}$$

$$x = 12 \text{ (reject)} \quad \text{or} \quad x = \frac{10}{3}$$

Max volume $V = \frac{19600}{27}$ when $x = \frac{10}{3}$.

Quick Reference — Key Formulas

Topic	Formula
Slope	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$
Circle	$(x - h)^2 + (y - k)^2 = r^2$
Point-Slope	$y - y_1 = m(x - x_1)$
Slope-Intercept	$y = mx + b$
Derivative	$f'(x) = \lim_{h \to 0} \dfrac{f(x + h) - f(x)}{h}$
Power Rule	$\dfrac{d}{dx}[x^n] = nx^{n-1}$

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Product Rule	$(fg)' = fg' + gf'$
Quotient Rule	$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$
Chain Rule	$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
sin limit	$\lim_{x \to 0} \frac{\sin x}{x} = 1$
Continuity	$f(c)$ defined, $\lim f(x)$ exists, $\lim f(x) = f(c)$
IVT	f continuous on $[a, b]$, C between $f(a), f(b) \Rightarrow \exists x : f(x) = C$
Extreme Value Thm	f continuous on $[a, b] \Rightarrow$ abs max and min exist

End of MTH 101 Notes (Lessons 01–23)

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